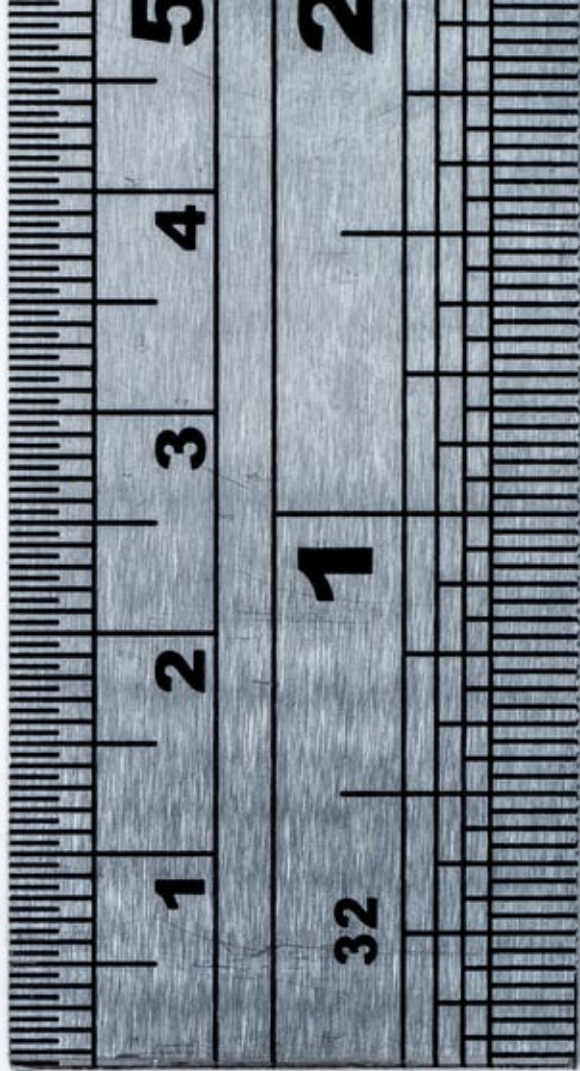
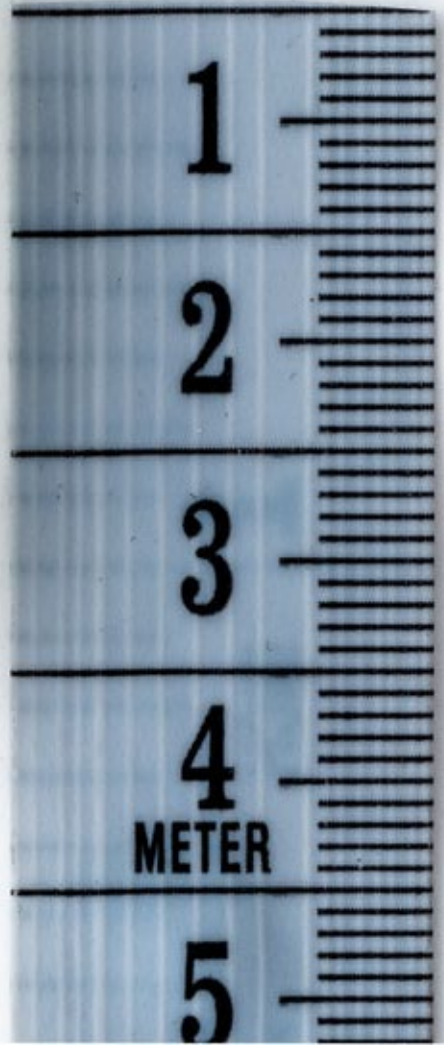




ME 300:
Engineering Thermodynamics

Dimensions

Primary dimensions and SI units

1. Mass - $M \rightarrow \text{kg}$

2. length - $L \rightarrow \text{m}$

3. time - $t \rightarrow \text{sec}$

4. temperature - $T \rightarrow \text{K}$

5. amount of matter - $N \rightarrow \text{moles}$

6. current - $I \rightarrow \text{Amp}$

7. luminosity - $C \rightarrow \text{candela}$



Derived dimensions and dimensional analysis

Pressure - P - Pa

Volume - V - m^3

Velocity - v - m/s

density - ρ - kg/m^3

$$P = \frac{F}{A} = \frac{Ma}{A}$$

$$Pa = \frac{N}{m^2} = \frac{kg \, m/s^2}{m^2} = \frac{kg}{m \cdot s^2}$$

Prefixes

kilo - 10^3

mega - 10^6

Centi - 10^{-2}

milli - 10^{-3}

micro - 10^{-6}

$$1 \text{ bar} = 100,000 \text{ Pa}$$

$$= 100 \text{ kPa}$$

$$= 0.1 \text{ MPa}$$



If 1 bar pressure is equal to 100,000 Pa, how many MPa is that?

- a) 100
- b) 1
- c) 0.1